AYUSH GUPTA

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2K23/EC/251

EDUCATION

B.TECH(Electronics and Communication Engineering)	2023-2027	Delhi Technological University, New Delhi	8.31
JKBOSE (Class XII)	2023	Dogra Higher Secondary School, Jammu, J&K	95.8 %
JKBOSE (Class X)	2021	S.D. Tara Puri Public High School, Jammu, J&K	87 %

ACADEMIC PROJECT

UART Communication System – **GITHUB** | Icarus Verilog + GTKWave, Python (Cocotb)

- Designed a UART module in Verilog to convert parallel data to serial and vice versa.
- Built a Python-based testbench using Cocotb for automated verification with randomized test vectors.
- Demonstrated deep understanding of state machines, baud rate configuration, and serial communication protocols.

FPGA-based Electronic Voting Machine (EVM) – **GITHUB** | AMD Vivado, Verilog HDL

- Implemented a 3-stage voting system (voter count setup, secure voting, result validation) using FSM logic.
- Synthesized and simulated the design on AMD Vivado with test cases verifying correct candidate selection and vote tallying.
- Showcased secure, efficient election logic design suitable for real-world FPGA-based applications.

Cache Memory Controller (Direct-Mapped) – **GITHUB** | Icarus Verilog, GTKWave.

- Designed and simulated a cache controller that interfaces between a CPU and main memory.
- Implemented tag matching, hit/miss logic, and cache fill operations using FSMs.
- Demonstrated knowledge of memory hierarchy, performance bottlenecks, and optimization techniques.

TECHNICAL SKILLS

Hardware Description Languages: Verilog, VHDL **Programming & Scripting:** C, C++, Python, MATLAB	**Simulation & Debugging Tools:** Icarus Verilog, GTKWave, Cocotb, AMD Vivado, Tinkercad, Arduino IDE, MATLAB, ModelSim, **Version Control:** Git, GitHub	**Embedded Systems:** Arduino, ESP32, UART Protocol, Serial Communication	**Digital Design Concepts:** Finite State Machines (FSMs), Memory Design, Cache Controllers, MUX/DEMUX, Verilog Testbenches, Protocol Simulation

OTHER INFORMATION

- LinkedIn: https://www.linkedin.com/in/ayush-gupta-/
- GITHUB: https://github.com/Ayush-Gupta-dev
- PORTFOLIO: https://portfolio-nine-sigma-74.vercel.app/